MTA
AUDIO EDITOR



User Manual

Complete guide to using MTA Audio Editor

MTA Audio Editor 0.3.0-r210

Build: 20261015171900

Alessandro De Salvo
braket71@gmail.com

https://github.com/desalvo/mta-audio-editor

EUPL-1.2

Digital Audio Workstation with MTA support

Windows macOS Linux Android iOS

MTA Audio Editor is a true **multitrack Digital Audio Workstation (DAW)** and does not require a project to be constrained to an MTA format. The **Multitrack DAW** project format is intended for unrestricted editing, mixing, effects and mastering; MTA8 and MTA16 remain available as dedicated project formats and export targets.

MTA means **Multi Track Audio**: a multitrack container that keeps musical parts separately addressable during playback. MTA Audio Editor can import and edit existing files, but it can also **create new MTA8/MTA16 files** from tracks or stems and export the completed project as **MTA, WAV, MP3, MP3+G or FLAC**; individual tracks can be exported separately as WAV, MP3 or FLAC.

This manual covers the complete MTA Audio Editor workflow: access and platforms, projects, import, waveforms, editing, synchronization, mixing, effects, musical metadata, stem separation, MTA8/MTA16 export, desktop clients and mobile apps. Procedures include screenshots, diagrams and practical examples.

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1. Quick start

Start by choosing how the project is created: empty, imported from an existing MTA, built from individual audio files, or generated from a stereo mix through stem separation. If the final destination is MTA, also choose MTA8/MTA16 and the target device profile.

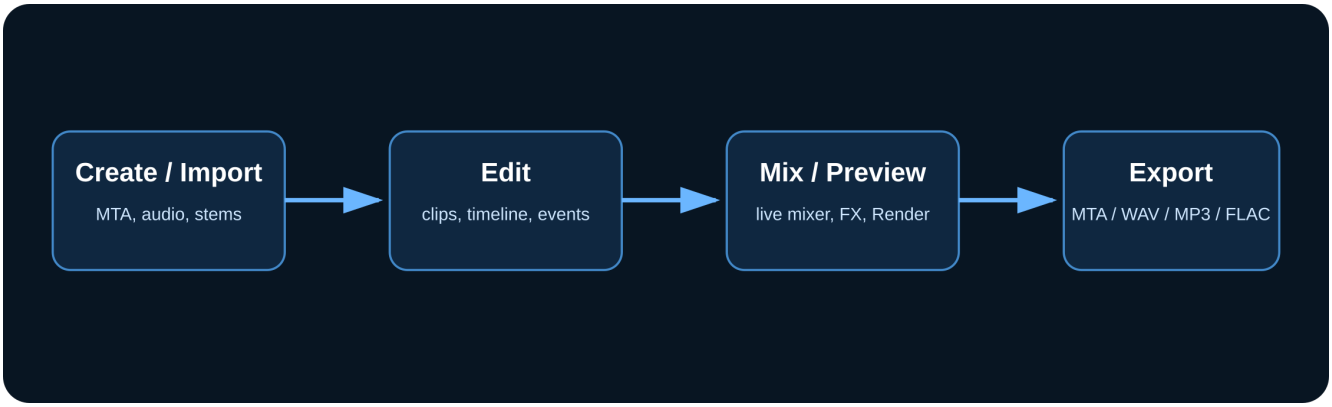


Figure 1 - Project lifecycle. Source audio remains available while clips, events and processing are edited non-destructively.

1. **Open or create.** Use New project, Import MTA or Open project.
2. **Inspect tracks.** Check names, roles, synchronization, channels and waveforms.
3. **Edit and mix.** Work with the timeline, faders, pan, mute/solo and inserts while playing.
4. **Check Render.** Compare dynamic preview with the rendered master.
5. **Export.** Select the device profile and verify Click/Melody and slot mapping.

2. Platforms and operating modes

The project model and editor are shared across platforms; file handling differs by client.

Platform	Scenario	Files	Processing
Browser / server	Centralized and multi-user operation	Server workspace	Server-side FFmpeg, waveforms, stems and renders
Docker / Kubernetes	Production and team deployments	Persistent volume / PVC	Server-side
Windows / macOS	Local desktop workflow	Native file pickers and bound project archive	Integrated local backend
Android / iOS / iPadOS	Mobile access to a server	System file pickers and sharing	Heavy operations remain on the server

Mobile first launch: the apps automatically use their preconfigured service. They do not ask for a server address and never display the built-in default address. Server settings can later store a custom address; only a user-configured custom value may be displayed and edited.

3. DAW workspace

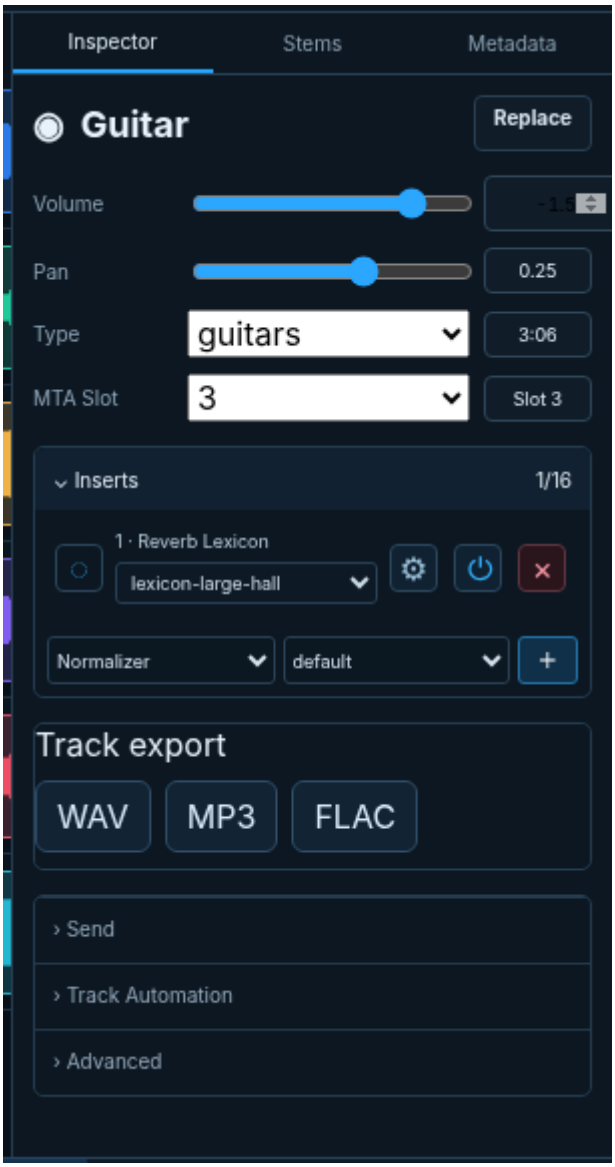
The main screen follows a DAW layout: transport at the top, timeline and waveforms in the center, inspector on the right, mixer at the bottom, and musical event/import/export functions around the workspace.



Figure 2 - Real application workstation captured from the current build.



Real application view - Transport: start, rewind, Play/ Pause, tempo, Render, Follow and zoom.



Real application view - Inspector: track properties, level, pan, type, MTA slot and insert chain.

3.1 Main panels

The **Timeline** manages clips and time selection. The **Inspector** edits the selected track. The **Mixer** provides a compact view of tracks and master. The larger vertical fader is the volume control; adjacent meters are reserved for realtime signal level, so no redundant static level bar is shown. **Lyrics/Chords/ Markers** manage timed musical events.

3.2 Selection and state

The selected track provides context for insert editing, import/replace, split and single-track export. Mute, Solo, level, pan and FX state remain available in the mixer.

4. Projects, archives and autosave

The **Multitrack DAW** format is the recommended default for general editing and mixing: it does not constrain the project to 8 or 16 tracks. WAV, MP3, MP3+G and FLAC export directly; when MTA export is selected, MTA8 or MTA16 is chosen and slot mapping is applied only for that output.

A project stores tracks, clips, waveform previews, events, inserts, mix settings, target format, MTA profile, view state and metadata. Source files and required attachments remain in the project workspace.

1. **New desktop project.** Choose a name and save destination; the local archive is bound to the project.
2. **Desktop MTA import.** A project destination is selected before import starts. Cancelling the dialog cancels the import.
3. **Autosave.** Changes are automatically persisted when enabled; manual Save remains available.
4. **Project Info.** Project/File - Project Info shows name, project ID, home/workspace, location, target, duration, track count, artist, BPM, key and MTA profile.

Project archive versus MTA export: the project archive contains editing information that does not necessarily belong in the final MTA.

5. MTA and audio import

MTA import creates editor tracks from the available audio streams, reads usable names/roles and supported attachments, and keeps the original source. Audio import can add a new track or replace the selected track.

Operation	Use	Result
Import MTA	Start from an existing MTA	Project with available streams and metadata
Import Audio Track	Add a stem or part	New project track
Import & Replace	Replace one part while preserving role/structure	New source on the selected track
Import & Separate	Start from a stereo mix	Demucs job and generated stems

Example: select Guitar, choose Import & Replace, check the entry point and Solo it with drums and bass. Fine-tune any offset by moving the track or editing its clip.

6. Waveforms and cache

Waveforms are graphical previews, not the audio source. They are calculated when required, stored in the project with a revision and loaded immediately when a project is reopened. Changes that invalidate the preview trigger recalculation.

> **Important:** playback does not depend on waveform availability. Audio can play while waveform generation is still in progress.

7. Transport, buffering, synchronization and Follow

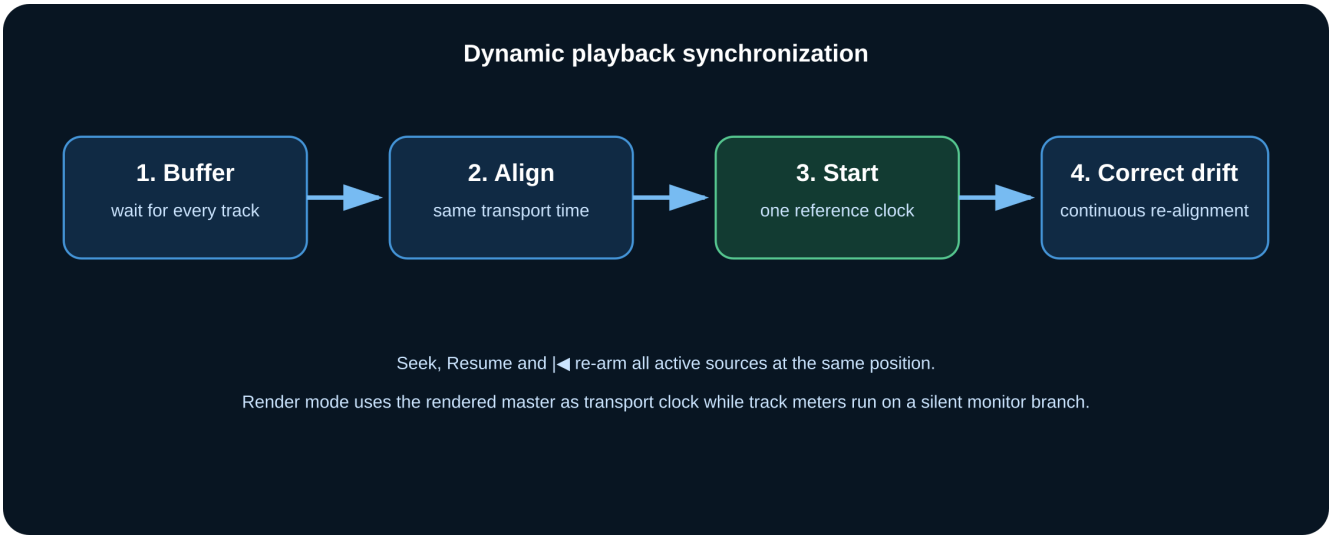


Figure 3 - Dynamic preview startup and synchronization.

The |◀ control returns the transport and active sources to 00:00.000. Rewind/forward change position; Play starts or resumes; Pause keeps the current position; Stop closes the current preview session.

7.1 Dynamic preview

Before playback, all tracks are prepared and buffered. The UI reports **Buffering audio... please wait** and then synchronization. Playback starts only when sources are ready. A reference clock monitors drift and realigns active sources when necessary.

7.2 Render mode

Render plays a backend-generated master preview. Browser-live parameters remain immediate; changes that are baked into the master trigger an automatic refresh at the current position without a manual Stop/Play cycle.

7.3 Follow

Follow keeps the playhead visible in both dynamic and Render modes. In Render mode the rendered master explicitly acts as the transport clock.

8. Non-destructive editing

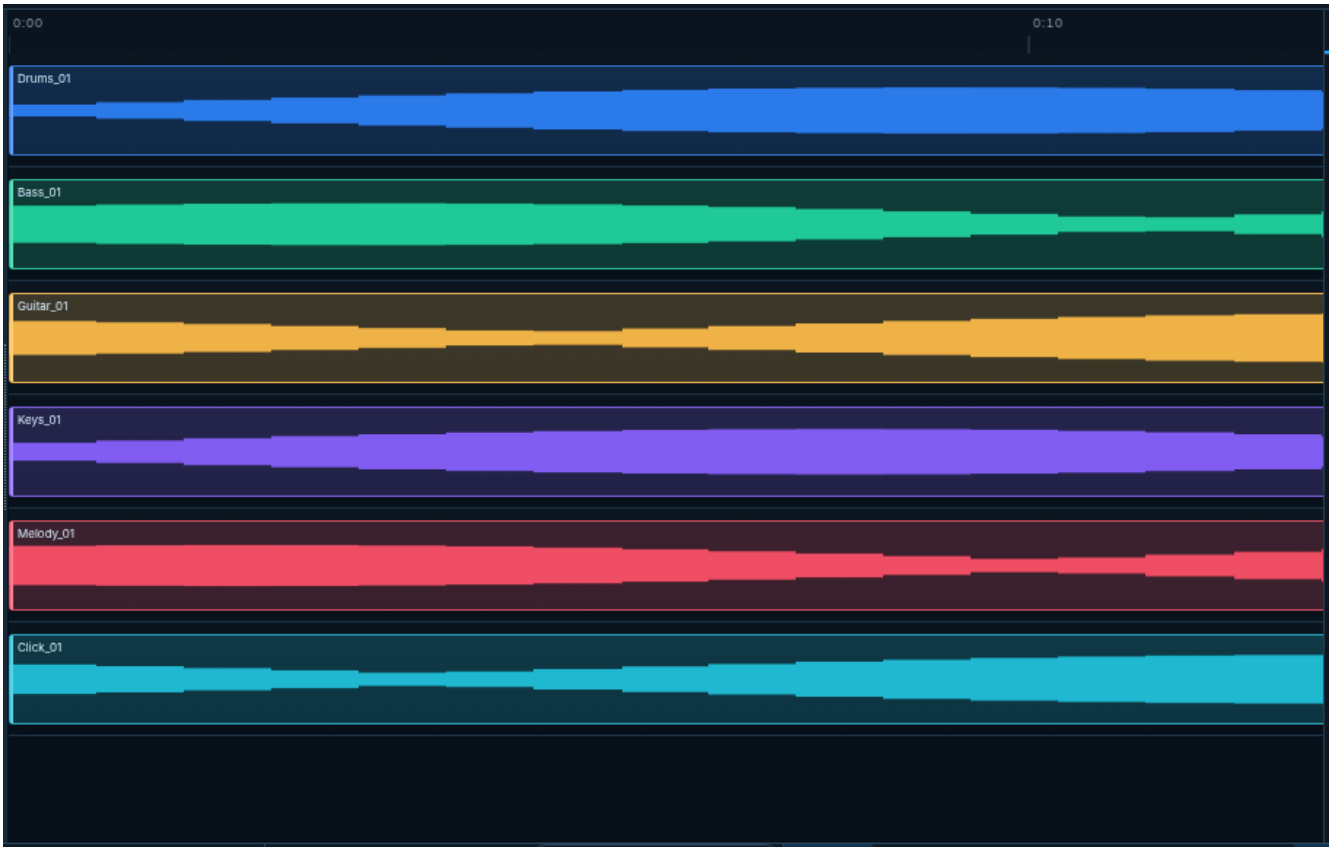


Figure 4 - Timeline and range selection.

Editing changes clip ranges and positions without rewriting the immutable source audio.

Command	Effect	Typical use
Split	Split a clip at the chosen point	Create independent sections
Remove	Remove a selected interval	Delete noise or a part
Ripple	Close the gap by shifting later material	Shorten a section
Delete song segment	Global cut plus event realignment	Remove a verse/section
Move track	Apply a global time offset	Correct alignment

9. Tracks, roles and MTA slots

Each track contains name, role/type, source, clips, channel information, volume, pan, mute/solo, inserts, waveform state and an **assigned MTA slot**. The Inspector exposes an editable **MTA Slot** field: **Auto** delegates placement to the device profile, while 1...8 or 1...16 stores the preferred output slot. The value is reused as the initial export mapping and is updated when the user confirms a mapping. Multiple tracks may share a slot when they must be mixed into one MTA output stream. Track roles remain important to Auto Mix and mapping; `click` identifies the metronome and `melody` the guide melody.

Click identification: explicit metadata and track name/role are preferred. Stream position alone is not sufficient because MTA8 device families use different Click/Melody orders.

Fixed and adaptive metronome

Metronome generates a click at the project BPM. **Adaptive** is intended for songs with accelerando, ritardando or internal tempo changes: every request reanalyses the complete current song from scratch, excludes any existing click track from detection, rebuilds a beat-by-beat tempo map and regenerates the click on the detected beats. A previous adaptive map is never reused.

The track context menu also provides **Metronome from this track** and **Adaptive metronome from this track**. In these modes rhythmic analysis uses only the selected track, which is useful for drums, percussion or other sources with clear transients. Fixed mode estimates BPM and beat phase; adaptive mode rebuilds the complete beat-by-beat tempo map from the selected reference track.

The adaptive metronome can be moved without reanalysis using **Adaptive shift** in milliseconds or beats. The transport can display and accept either absolute `mm:ss.mmm` time or musical `bar:beat` position; conversion uses the fixed BPM/phase grid in fixed mode or the detected beat map in adaptive mode.

10. Live mixer and VU meters



Figure 5 - Mixer and meters.

Track level, pan, mute/unmute, solo/unsolo and FX state can be changed while playback continues. Global Mute/Unmute, Solo/Unsolo and FX Bypass/Enable controls simplify mix comparisons.

RealTime meters monitor both tracks and master. In Render mode the per-track analyser path is separated from speaker output, so the master can remain the only audible stream while track meters still receive the real signal.

> **Balancing example:** start playback, establish drums as a reference, balance bass and instruments, use Pan to create space, then check the master. No Stop/Play cycle is required for these changes.

11. Insert FX, presets and controls

Tracks and master support chains of up to 16 inserts. Available processors include normalization, compression, limiting, Delay, Lexicon-style Reverb, Room/Ambience, 32-band Graphic EQ, Amplify, Stereo Imager, Maximizer/Loudness, Mastering Wizard, De-Noise and Crackling Cleaner.

Operation	While playing
Bypass / Enable	Applied automatically
Preset change	The affected preview is refreshed
Custom parameters	Persisted and applied
Add / remove insert	The chain updates without manual Stop/Play

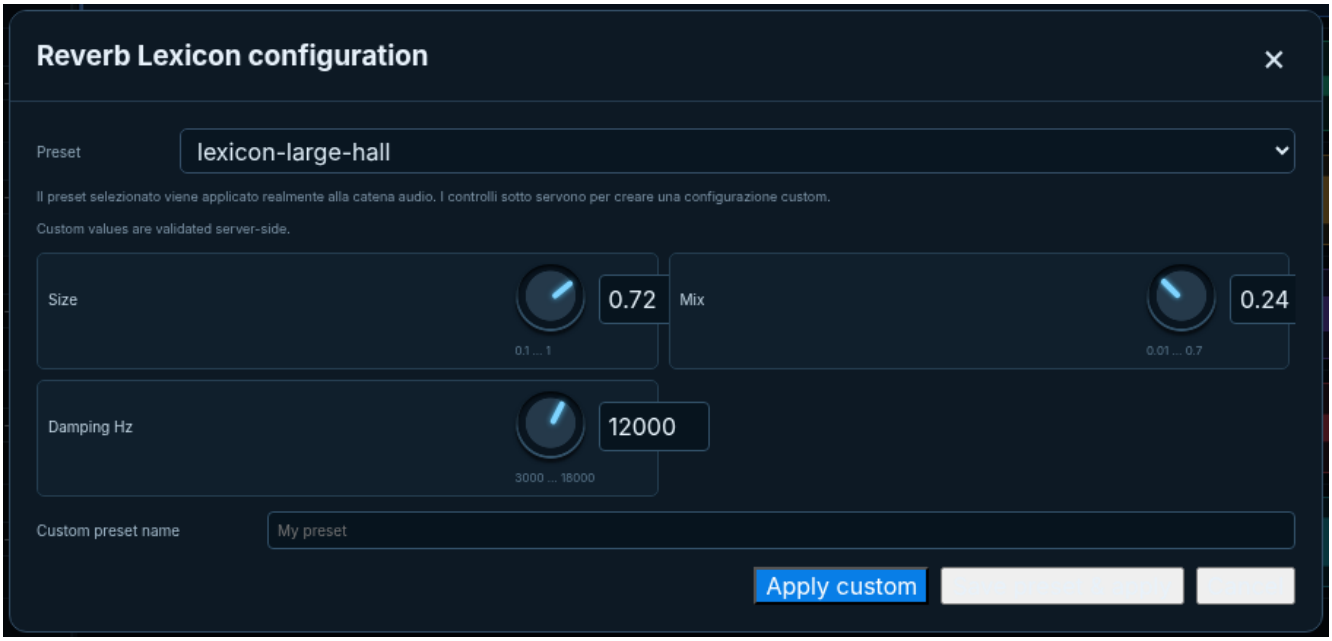


Figure 6 - Real plugin editor: preset selection immediately refreshes the visible parameters and continuous values expose synchronized knob and numeric controls.

Changing a preset updates the plugin's visible parameters **immediately** to values appropriate for that preset: for example Slapback selects a short delay, Large Hall increases reverb size/mix, Wide increases stereo width and Strong raises noise/crackle reduction. Continuous parameters use a synchronized **rotary knob plus numeric input**; moving either control updates the other. A knob commits the value when the adjustment is released; a numeric input commits directly when **Enter** is pressed or when focus leaves the field. The 32-band graphic EQ keeps its dedicated faders. Factory presets can be complemented by user presets, and all parameters are validated by the backend.

12. Auto Mix and master

Auto Mix applies deterministic, role-aware balancing rules to track gains, pan and processing. It is reversible: enabling it stores a snapshot and disabling it restores that previous state.

Recommended workflow: use Auto Mix as a starting point, then manually refine the result while checking guide vocal/melody, Click, master dynamics and the densest sections of the song.

13. Lyrics, Chords and Markers

Lyrics, Chords and Markers are timed project events. Lyrics can be extracted with Whisper large-v3 with word-level timing; chords prefer Chordino/NNLS-Chroma when available and fall back to the internal harmonic analyser. They remain editable, are saved in the project and MTA exports, can be downloaded as text, and can be rendered to PDF with a selectable chord color.

> **Example:** add Intro, Verse, Chorus and Solo markers, timed chord symbols and lyric lines. A global song cut shifts later events together with the audio.

14. Stem separation

Import & Separate uses Demucs and can optionally extract synchronized lyrics and chords in the same workflow. For higher accuracy, lyrics extraction uses the isolated Vocals stem when available, while chord analysis uses the original mix. Stems, lyrics and chords are saved into the project and carried into MTA exports.

1. Select the stereo source.
2. Choose Multitrack DAW, MTA8, MTA16 or the current project.
3. Wait for server progress.
4. Rename and classify generated stems.
5. Add Click/Melody if available and synchronize.

15. MTA8/MTA16 and device profiles

MTA export for Merish5 / Xynthia2: files use the .mta extension; MTA8 identifies the number and order of slots, not a filename extension. Select the Merish5 / Xynthia2 profile (Melody 7, Click 8). Export uses 44.1 kHz stereo audio at 320 kbit/s with equal stream lengths. Loading on the device remains unverified: proprietary layout and attachments differ from stock samples. A filename extension alone does not guarantee playback.

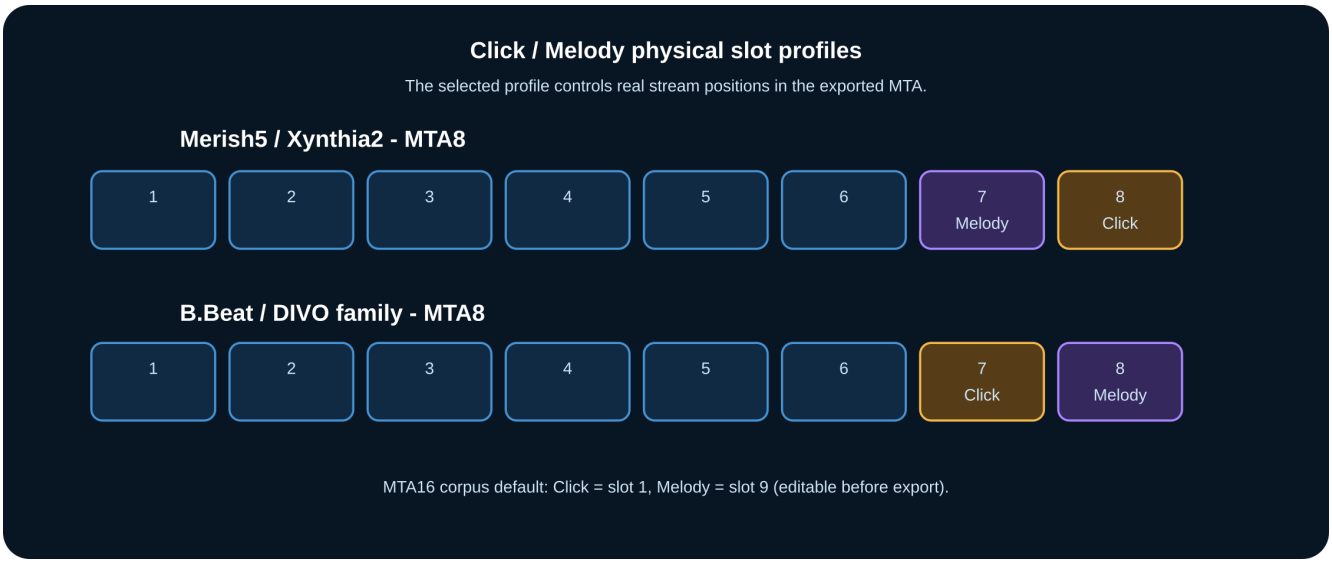


Figure 7 - Physical Click/Melody positions used by export profiles.

MTA (Multi Track Audio) is a multitrack format containing audio streams plus metadata/attachments. Device-specific export must preserve the real stream position, not only an internal logical slot label.

Profile	Click	Melody	Behavior
Merish5 / Xynthia2 - MTA8	8	7	Physical positions preserved
B.Beat / B.Beat Plus / Evo / Pro16 / DIVO - MTA8	7	8	Physical positions preserved
M-Live MTA16 - corpus default	1	9	Suggested and always editable
Generic MTA	Manual		No device-specific assumption

MTA16: the available MTA16 samples consistently place Metronome in stream 1. Explicit Melody Track appears most often in stream 9 and in one sample in stream 10, so slot 9 is a practical default rather than a universal requirement.

16. MTA creation, export and slot mapping



Figure 8 - Real Export panel.

The application can **create a new MTA8 or MTA16 from scratch** as well as modify an imported MTA. A completed project can be exported as MTA, WAV, MP3, MP3+G, FLAC or karaoke MP4, while each individual track can be exported separately as WAV, MP3 or FLAC. MP4 export is enabled only when synchronized lyrics exist and can use either a black background or an optional JPG/PNG/WEBP image; word-level timing is used for karaoke highlighting when available. For MTA, select the device profile and verify slot mapping. When a project contains more tracks than the MTA capacity, multiple project tracks can be assigned to one output slot and mixed before container generation.

1. Choose MTA8 or MTA16.
2. Select the target device profile.
3. Check Click and Melody roles.
4. Review or customize slot mapping.
5. Export and test the result on the target device/firmware before live use.

Silent slots: when needed, export creates silent intermediate streams to preserve a physical position. Without them, an item assigned to slot 9 could become the fourth real stream.

17. Android and iOS/iPadOS

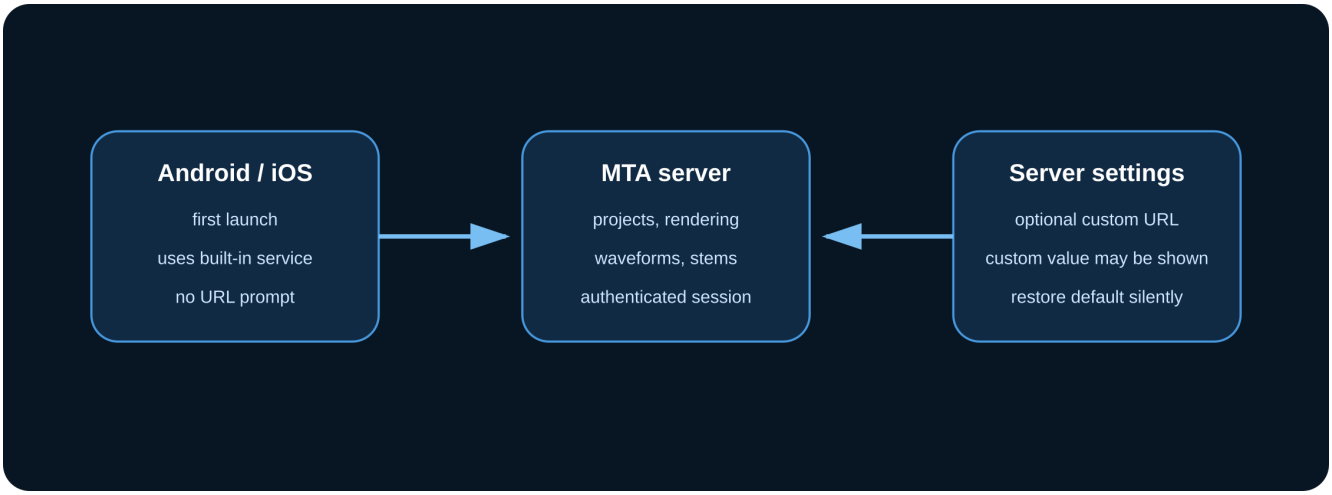


Figure 9 - Mobile connection and optional custom server configuration.

The mobile apps are native shells for the server UI. On first launch they connect automatically to the preconfigured service: installation and first launch never ask for a URL, and the built-in default address is never displayed.

Open **Server settings** only when using a different server. The input can display the custom value chosen by the user; selecting **Use default** removes that custom value and returns to the built-in service without exposing its address.

Function	Android	iOS/iPadOS
Import	Storage Access Framework	Document Picker / Files
Export	Create Document	Files / Document Picker
Sharing	Sharesheet	Share Sheet
Session	WebView cookies	WKHTTPCookieStore

Offline mode: the mobile clients do not bundle full FFmpeg/PyTorch/Demucs runtimes. Rendering, stem separation and MTA generation require a reachable server.

18. Windows and macOS

Desktop clients run the local backend and native WebView. File choosers are provided by the operating system. macOS packages target both Intel and Apple Silicon; Windows can be packaged with an installer.

A local project archive binding is persisted across restarts. Save As can change its path. Native import/upload limits are configurable in Settings.

19. Accounts, TOTP, security and sharing

Users, passwords, TOTP, email verification, roles and sharing apply to the online/server edition and to mobile apps. Mobile apps connect to a cloud/remote server because heavy operations - FFmpeg rendering, MTA generation, waveform work and Demucs separation - run server-side; therefore they use the server account. In local native/desktop single-user mode authentication features are unnecessary and hidden, and processing/files stay on the user's computer.

Recommendation: public servers should use HTTPS, strong passwords, TOTP where appropriate, regular persistent-volume backups and upgrades validated through the production gate.

20. End-to-end examples

20.1 Prepare an MTA for a show

1. Import the MTA and save the project.
2. Check tracks, roles, Click/Melody and the device profile.
3. Start Play, wait for buffering, and adjust mixer/pan/FX live.
4. Enable Render and compare the master while keeping Follow and meters active.
5. Review Lyrics, Chords and Markers.
6. Export the MTA and validate it on the target device.

20.2 Build an MTA from a stereo mix

1. Create a project and run Import & Separate.
2. Classify the generated stems.
3. Import or create Click and Melody.
4. Correct offsets, then mix and process.
5. Choose MTA8/MTA16 and the device profile.
6. Review slot mapping and export.

20.3 Correct a track while the song plays

Solo the target track, change pan/level, bypass an insert and compare presets, then restore Solo. These operations are applied without a manual Stop/Play cycle; Render mode refreshes the required preview automatically.

21. Quick controls

Action	Control
Return to start	|◀ transport button
Play / Pause	Transport Play/Pause
Undo / Redo	Standard platform shortcuts
Global Mute/Solo	All controls in mixer
Global FX bypass/enable	Global FX controls
Project information	Project/File - Project Info

22. Troubleshooting

Issue	Check / action
Play does not start immediately	Inspect the reported preparation stage and source format; cached compatible audio should start promptly. For C++ inspect underruns and network buffering.
Tracks appear out of sync	Seek, Resume or use |◀; active sources are also continuously drift-corrected.
Follow does not scroll	Ensure Follow is enabled. It is supported in Render mode.
Meters inactive in Render	Enable RealTime meters; rendered metering is supported.
Click routed to the wrong device output	Check track role, MTA profile and physical slot mapping.
Waveform missing	Wait for waveform generation; playback remains available.
Mobile app cannot connect	If a custom server was configured, review it in Server settings or return to the default service.
MTA rejected by hardware	Check MTA8/MTA16 target, profile, slot capacity and target firmware validation.

23. Glossary

Term	Meaning
Clip	A non-destructive time range referencing source audio.

Track role	Musical function used by Auto Mix and MTA mapping.
Click	Metronome track.
Melody	Guide melody track.
Dynamic preview	Current: client track buffers. C++ server: PCM stream with adaptive buffering.
Render preview	Backend-generated master preview.
MTA slot	Logical and, for profiles, physical stream position in the MTA.
Stem	Separated component of a mix.
Attachment	Embedded MTA data such as metadata or timed sections.

24. Stem count

Stem separation offers **Auto / 2 / 4 / 6 / 8**. Auto is the default. The current iPhone/iPad build performs Demucs on the server; 8 stems requires a compatible extended backend model.

On-device Demucs on iPhone/iPad. Auto mode uses Core ML on the device when a compatible local model is available and falls back to the server otherwise. Models can be downloaded once from the server and reused offline. The current conservative on-device limit is 12 minutes per source file; longer material is processed by the server in Auto mode.

25. Native updates and Demucs models

Native Windows, macOS, Android, iPhone and iPad applications check GitHub for newer versions. The **Stable** channel uses tagged releases only, while **Early release** also includes the rolling `early-main` prerelease produced by the latest successful main build.

On iPhone/iPad the app automatically provisions a default 4-stem Demucs Core ML model: it prefers a bundled validated model and otherwise downloads it from the configured server. SHA-256 fingerprints are compared with the server inventory and installed models are refreshed when the fingerprint changes. Once installed, the local model remains available offline.

On iOS/iPadOS application updates must still use TestFlight/App Store or another operating-system-authorized distribution mechanism.

26. Automatic mobile Demucs model updates

The Docker/Kubernetes server periodically refreshes Core ML (iPhone/iPad) and ONNX (Android) model repositories from an HTTPS manifest, verifies SHA-256 and replaces files atomically. Mobile clients check the catalogue every six hours and at startup. **Update on Wi-Fi only** is enabled by default, so periodic model updates download only over Wi-Fi unless the switch is explicitly disabled. The baseline 4-stem model is provisioned at the first available network opportunity.

27. Model-driven stem separation

Stem counts are no longer limited to 2/4/6/8. The list is generated from models actually available on the server or installed on the device and may include 10, 12, 16, or more stems. Multitrack DAW projects keep all tracks; MTA8/MTA16 constraints are handled only during export.

28. Dynamic Demucs model management

The server periodically refreshes the full configured model catalogue, including official Demucs models available in the runtime and Core ML/ONNX artifacts published by the manifest. Mobile and desktop clients download the selected model on demand when splitting is requested. Local options can delete, re-download, or force-update one or more models. Administrators can blacklist a model: managed server artifacts are deleted and the model disappears from the web/client catalogue until it is removed from the blacklist.

29. AI model authentication

Demucs downloads use revocable runtime tokens scoped to `models:read`, separate from administrator privileges. Mobile apps obtain and renew the token automatically after a valid session.

30. Synchronized Lyrics and Chords

Use a track's context menu and choose **Extract lyrics** or **Extract chords**. Detected events are remapped to the track's actual project timeline, so offsets, clips and cuts are respected. Lyrics and chords become independent project data and remain available if the source track is later edited.

Lyrics can be downloaded as plain text, synchronized text with chords, or as a **PDF**. The PDF places chords above lyric lines and, when available in Project Info, shows title and author/performer at the top. MTA export embeds a lossless synchronized-event representation so the data stays inside the file and can round-trip back into the project.

> **Lyrics extraction:** uses local/upstream OpenAI Whisper when the optional engine is installed. Chords are analysed locally by the built-in engine.

31. High-accuracy Lyrics/Chords and karaoke video

Import & Separate can extract lyrics and/or chords in the same workflow. Lyrics use Whisper in high-accuracy mode (large-v3 by default, beam search and word-level timestamps). Chords prefer Chordino/NNLS-Chroma when available and fall back to the built-in harmonic analyser.

The Lyrics + Chords PDF lets the user choose a chord colour. **MP4 Karaoke** export renders the project mix audio with synchronized lyrics; an optional JPG/PNG/WEBP background image can be used and chords can also be displayed above the lyrics. MP4 export is unavailable when synchronized lyrics are missing.

32. ChordPro and author-society repertoires

Synchronized lyrics and chords can also be exported as **ChordPro** (.cho): title, artist, writers, key and BPM are emitted as ChordPro directives, while chord symbols are aligned into the lyrics using word-level timestamps when available.

Project Info can store original title and writers/composers and provides **Search rights repertoires**. SIAE and Soundreef are selected by default. When an official/licensed structured endpoint is configured, candidate works are shown directly and one or more can be selected; when no documented public API is available, the app links to the official repertoire and allows a verified result to be recorded manually. Selected records (society, original title, writers/performers, publishers and identifiers such as ISWC/IPI/work ID) are stored in the project, embedded in MTA exports and printed at the end of Lyrics + Chords PDFs.

M-Live / Merish MP3 export

The output formats include **M-Live / Merish MP3**, producing a normal MP3 with synchronized lyrics, chords and markers in ID3v2.3/SYLT metadata. Title, performer and authors are embedded when available in Project Info. The technical profile is documented in `MTA-Audio-Editor-M-Live-Merish-MP3-Format-Specification-EN.pdf`.

Project sample rate and clip conversion

Every project has a **fixed sample rate** selected at creation time: 44.1 kHz (default), 48 kHz, or 96 kHz. It can be changed from Project Info only while the project has no active tracks. Audio imported at 44.1/48/96 kHz is retained in Originals while the timeline copy is automatically resampled to the project rate when required.

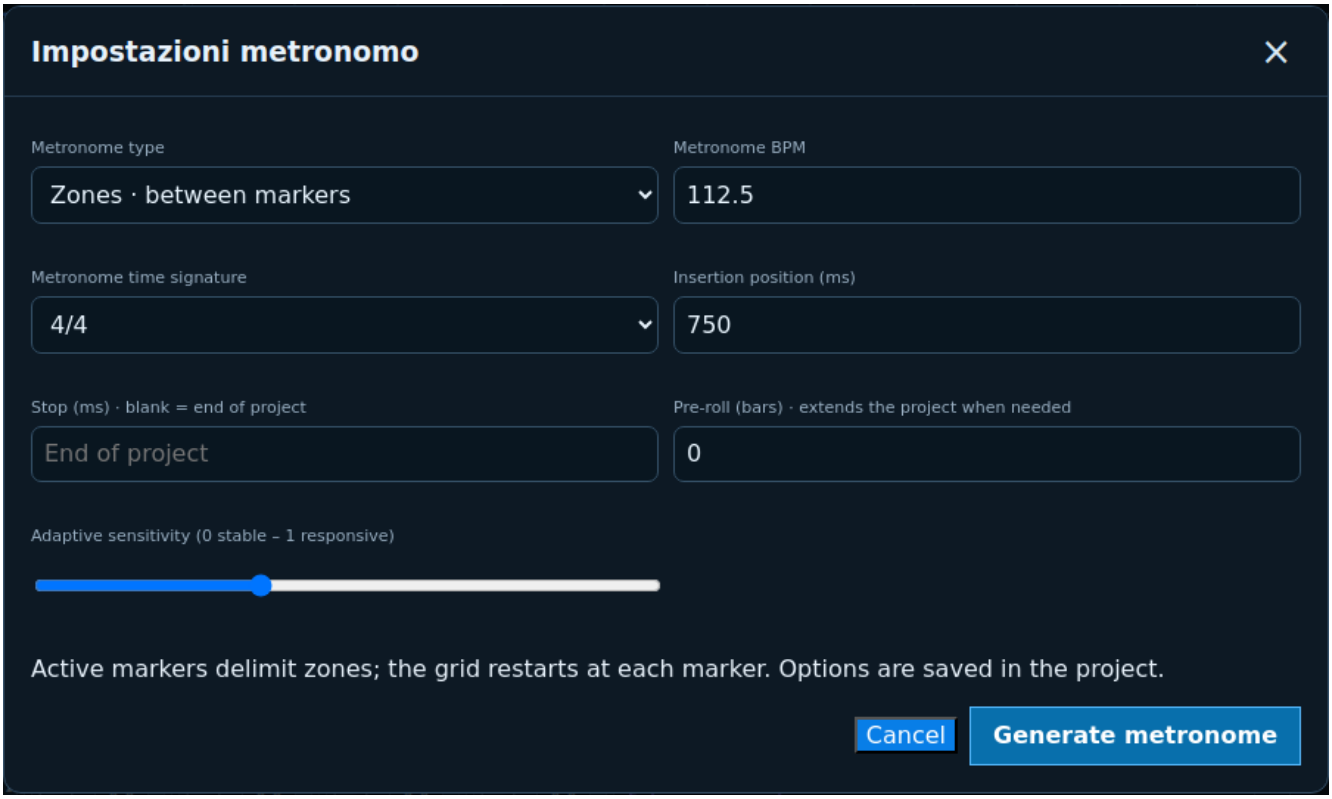
WAV and MP3+G export

WAV export supports **24-bit PCM** or **32-bit float** at 44.1, 48, or 96 kHz. **MP3+G** export creates a ZIP containing a same-basename MP3 audio file and CDG graphics file with synchronized karaoke lyrics and chords.

33. Zone metronome and personal tools

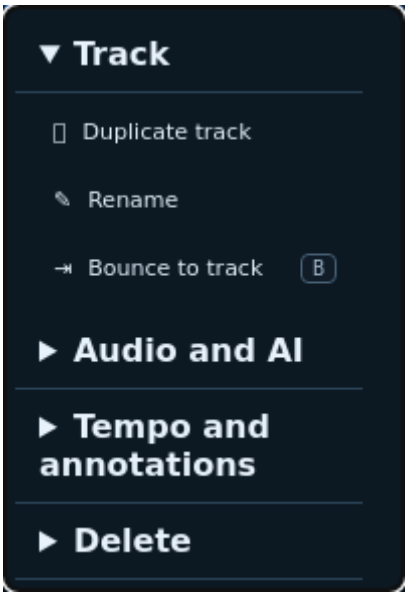
Insert a zone metronome at the current transport position from the context menu, or enter an insertion position in milliseconds in its creation dialog. The first beat matches insertion; subsequent markers restart the grid. Pre-roll adds 0-64 bars before insertion, extending the project at the beginning when

necessary. A Stop limit can mute the end. Moving, adding, disabling or removing a marker regenerates and saves the zone WAV, updating waveform, cache and playback. Manual zone BPM/meter settings follow marker identities.



Metronome creation: mode, independent grid, insertion and pre-roll.

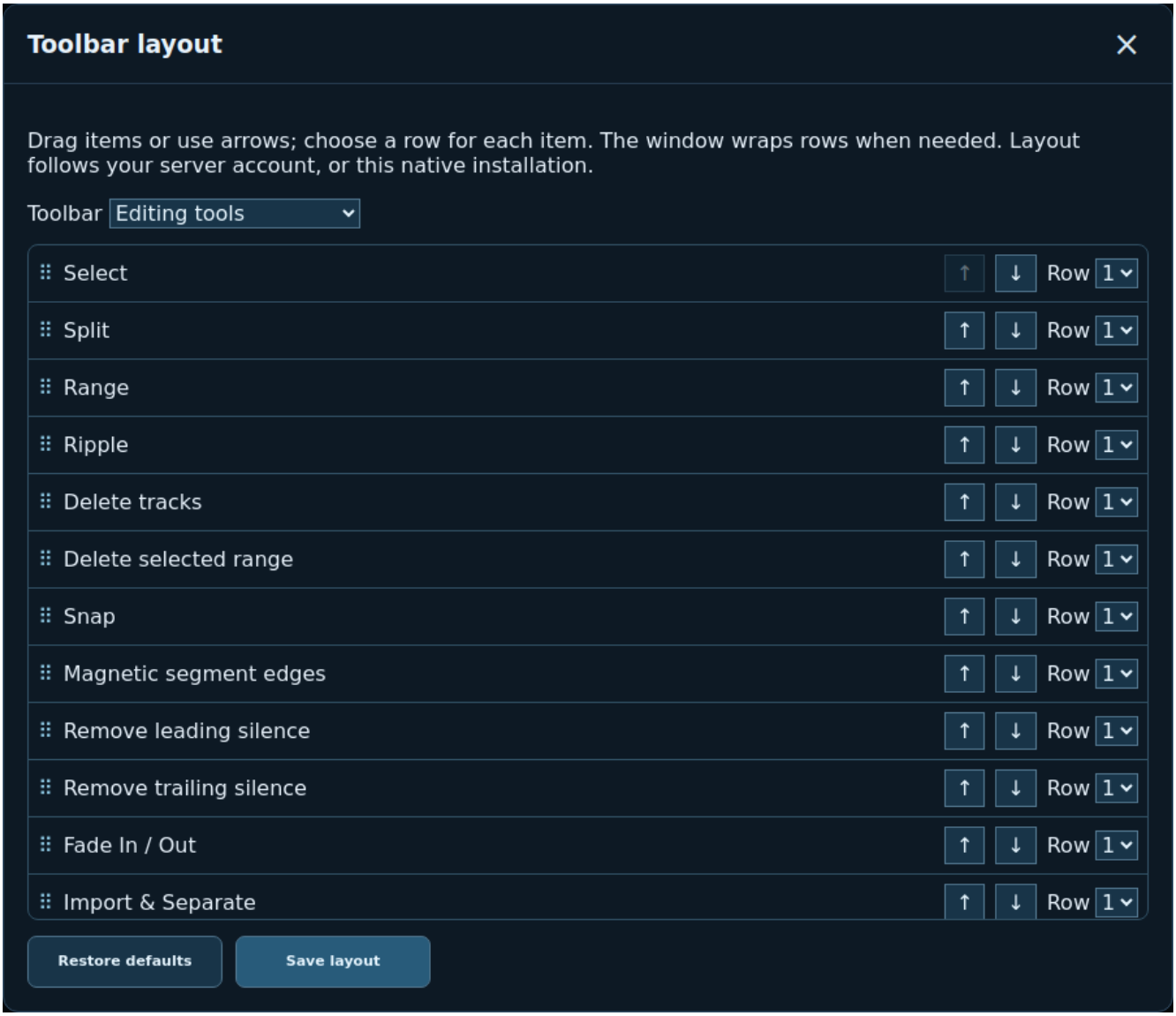
Audio BPM changes project playback speed; Grid changes only metronome BPM/meter, snap and musical display, retaining track speed. Adaptive metronome timing is derived from audio instead. Delay/offset also affects playback; units include milliseconds, beats, bars and bars:beats. R regenerates the selected metronome/chords track; B bounces a track. Snap and magnetic edges guide movement without introducing overlaps on the same track.



Categorized context menu: stays in the viewport and scrolls when necessary.

Toolbars and AI monitor

Horizontal toolbar buttons show icons; names appear in tooltips and remain accessible to assistive technology. The toolbar configuration button, also available in Settings, opens the item list: choose Transport and project or Editing tools, drag or use arrows to reorder, and assign each item to one of six rows. Save the layout; Restore defaults also resets monitor position. Rows wrap on narrow screens. Timeline zoom retains the time at the view center while dragging and after release; +/− buttons and presets use the same anchor, within project scroll boundaries.



Accessible reordering and row selection.

The AI monitor starts at the left margin, below navigation. Drag its dotted handle to move it; with the handle focused, arrow keys move it and Home resets it. Resizing keeps it in view. Server layout and position belong to the authenticated user; native apps store them in local preferences. Mixer Pan and Gain have dedicated labels; FX opens channel or master effects.



Updated mixer with Gain, Pan and FX.

34. Audio, VST3 and MIDI

The Current engine remains selectable. C++ DSP is available in native apps and on the server; server administrators choose it under Web App settings. Browser C++ streaming uses a dedicated PCM worker queue and AudioWorklet; native transport can use the local device. VU meters show measured levels. Complex projects or slow plugins can still cause underruns: inspect diagnostics and reduce load.

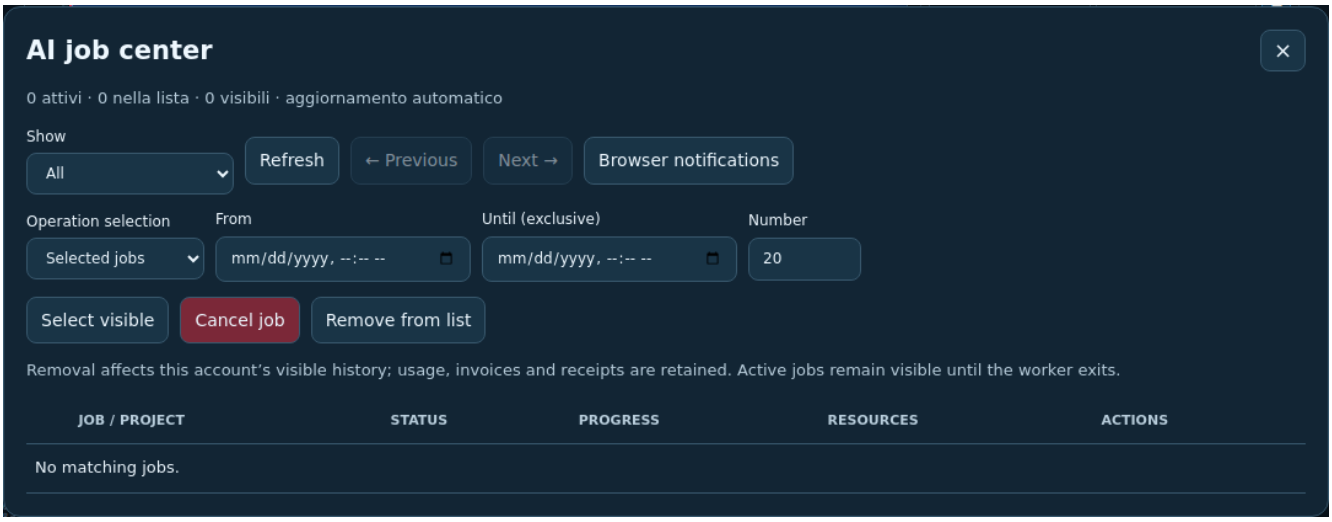
Bounce to track applies selected-track processing; Bounce project to track uses the master mix. VST3 plugins require binaries matching the OS/architecture of the processing computer, including remote use. The optional catalog provides installation, per-plugin progress and controlled uninstall; stored project presets are retained. Close active plugin instances before removing their package. The IT/EN VST3 manual details original editors, VSTi, MIDI and host synchronization.

Online VST3 manual · PDF VST3

35. AI, usage and subscriptions

Server AI engines use the global queue. Native apps use one token per remote instance, associated with a user. AI jobs exposes status, CPU/GPU, progress and cancellation even after closing the originating window. Completion notices appear when the origin is closed and results have been saved in the project. Configured SMTP enables job notifications and usage summaries. GPU estimates are identified as estimates; measurement errors must not be interpreted as zero usage.

CPU/GPU consumption accepted by the server is accounted for quotas and billing. Live counters remain separate until job completion and acceptance. Source evidence is retained for audit; missing or invalid data does not authorize charges.



AI job center: status, resources and background controls.

Profile Limits and Quotas collects storage, CPU/GPU, billable usage and history. Subscriptions and credits are available only when enabled by the administrator for that user. Credits cover active-subscription overage, not use without a subscription. An admitted job completes even if its credits run out, without charges beyond its admitted budget. A job requiring overage credits is rejected when they are zero: purchase credits if enabled or contact an administrator. PayPal requires server configuration; it does not imply automatic PayPal renewal.

34. Advanced MIDI and AI studio checkpoint

Open the MIDI sidebar to create/import independent tracks, export multitrack MIDI, create a linked MIDI metronome or open its tempo graph. Sidebar state is saved for the current identity through lazy database patches.

Select multiple notes, move/resize them or use quantize, swing, humanize, split/join and transpose. Musical changes support undo/redo. The searchable paginated event list includes notes, CC, program, bend, pressure, SysEx and metadata. Controller lanes support CC, 14-bit bend, pressure and program events, shared zoom/scroll, group dragging, arrow keys, snap and undo/redo. Save MIDI before closing. The piano roll draws the visible viewport rather than allocating a full-length bitmap.

Import/export supports SMF0/1/2 and SMPTE 24/25/29.97/30. SMF2 retains independent pattern tempo maps; SMF0/1 preserve timestamps using the project grid. Combining independent patterns may require channel reassignment. UMP uses its lossless JSON export.

FluidSynth with the reviewed FluidR3 GM bank is an optional per-track instrument with program, bank, gain, polyphony and reverb. Missing instruments do not prevent MIDI editing. The bank and library require separate installation. A linked MIDI metronome follows marker/grid changes: unlink it before manually editing notes. The tempo graph edits decimal BPM through drag, arrows or double-click without stretching audio. Adaptive timing is read-only.

The MIDI sidebar opens AI Vocal Studio and Audio-to-MIDI. Select a MIDI melody for 2–8 editable harmony parts, or up to 32 independent pitched audio sources for Basic Pitch CPU transcription. Source clips, delay and region offsets are respected. Tabs expose source, parameters, execution and results. Close the window without cancelling; find the job in All jobs. Results are saved atomically when the project is unchanged; otherwise review and import explicitly. Retry can reuse unchanged completed sources.

Singing synthesis is not available: authorized voicebanks, lyrics/phonemes, multimodel transcription, dedicated ASIO and complete hardware routing remain open. See docs/STUDIO_WORKFLOWS_R210_EN.md.

YourMT3 noPS is an additional operator-installed CPU backend for full-mix transcription into predicted GM/percussion parts. A real piano fixture passed through the queue and was saved as editable MIDI. This is not a ten-instrument accuracy benchmark. Velocity is generated; no calibrated confidence is reported. Installation, pinned hashes and licenses: docs/YOURMT3_RUNTIME_R210_EN.md. Vocal synthesis and multimodel reconciliation remain open.

Native Audio devices / ASIO / routing selects the host API, physical devices, buffer and stereo-master/input matrices. The compiled callback does not call Python. A duplex stream supports playback and stereo recording. Physical ASIO certification, independent per-track hardware buses and the ASIO corresponding-source redistribution bundle remain open. Native Studio models use dedicated local API paths; remote AI workflows use the shared token and recover results without duplicate charges.



MIDI controller lanes: shared editable events and undo/redo

Technical appendices

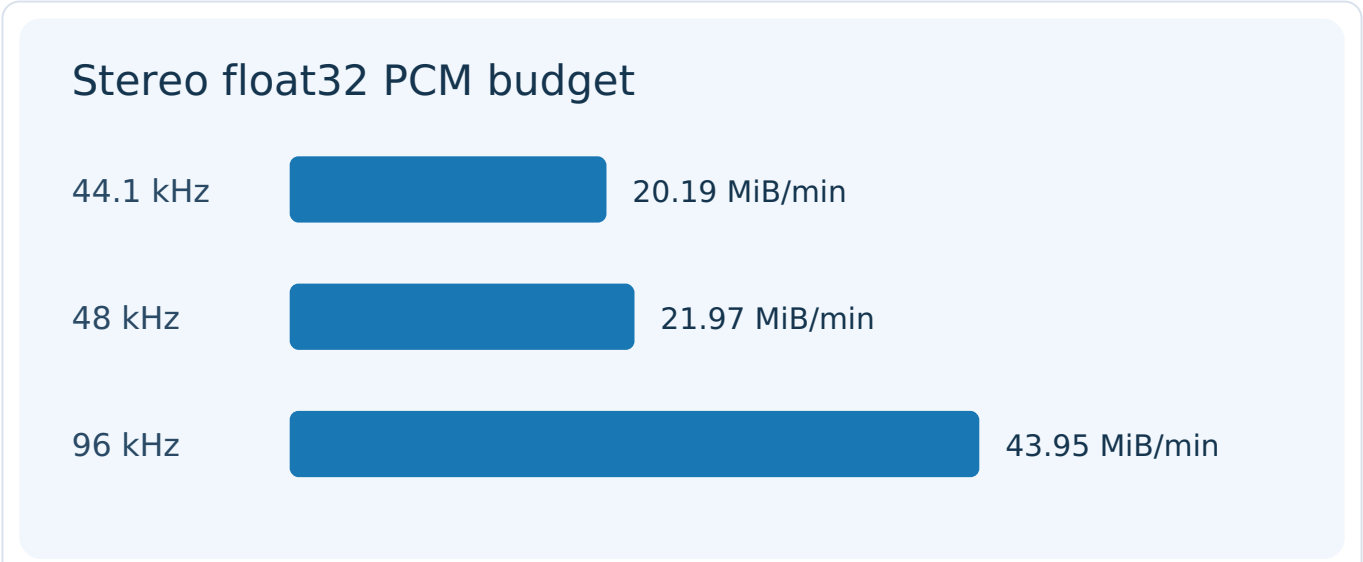
Documented state: 0.3.0-r210. Sizing guidance is an operational estimate, not a certified minimum. Local results do not certify physical devices, plugins or production servers.

A. Application types and capabilities

Type	Audio / DSP	AI	Plugin / MIDI
Web + server	Browser Web Audio; server C++ DSP	Global server queue; tokens for remote clients	Server Linux VST3, remote editor; Web MIDI where supported
Windows desktop	Current or C++; device transport	Local engines or token-authenticated remote server	Compatible Windows VST3; native MIDI
macOS desktop	Intel / Apple Silicon; Current or C++	Local models or server; accelerator support depends on engine	Matching-architecture macOS plugins; native MIDI
Linux desktop	Current or C++; local device	CPU / CUDA / ROCm depending on runtime and hardware	Linux VST3; ALSA/MIDI subject to availability
Android / iOS / iPadOS	Mobile shell; browser/ WebView audio	Server; on-device paths require a compatible model/ runtime	Server-side plugins; browser MIDI depends on platform

B. Resources and sizing

Profile	CPU / RAM	Disk / network	Note
Browser	4-8 GiB device RAM: planning baseline	HTTPS; stable network for streams and remote editors	256 MiB PCM cache; no guarantee for obsolete WebViews
Desktop / DSP	4 cores / 8 GiB starting point; 16+ GiB for dense sessions	SSD recommended; allow sources, exports and models	Plugins, sample rate and channel count determine actual load
Server editor	4 vCPU / 8 GiB for small tests; size for concurrency	Persistent volume, backups, bandwidth for concurrent sessions	Not a guaranteed user capacity
Server / desktop AI	16-32+ GiB RAM initial budget; model dependent	Separate weights/runtimes: several GiB, sometimes tens	VRAM/unified memory and precision depend on model; validate each engine
Mobile	Browser memory/CPU; avoid oversized projects	Stable Wi-Fi recommended for audio and remote plugins	Server DSP does not become phone-local



Theoretical PCM memory per minute, stereo float32 (excluding graphs, models and duplicate caches).

```
PCM bytes = seconds × sample rate × channels × 4
```

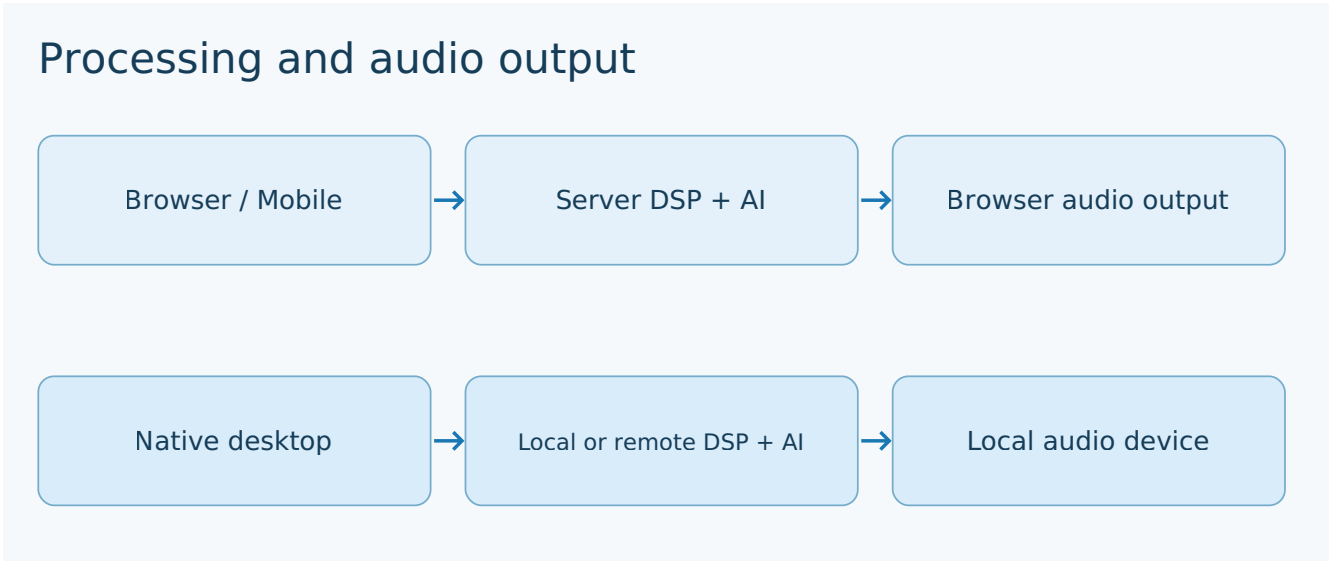
C. Limits and behavior

Area	Limit / behavior
Project	44.1 / 48 / 96 kHz; sample rate fixed after first track. Timeline up to 24 h.
Metronome	Standard, marker zones, adaptive. Pre-roll 0-64 bars; grid BPM 20-400 including decimals.
FX	Up to 16 inserts per chain; validate dependencies and VST3 compatibility per plugin.
MIDI	SMF0/1/2 up to 8 MiB; PPQ and SMPTE timebases. UMP/MIDI 2.0 depends on driver/provider, not guaranteed for every device.
AI	Global queue; configurable artifact limits and timeouts. An installed model does not guarantee GPU compatibility.
Streaming C++	Adaptive buffer 0.35-1.2 s plus block; not a guarantee of professional low-latency monitoring.
Storage / billing	New users default to 5 GiB; admins have no default quota. Quotas/credits depend on plan and permissions.

D. Observed performance and validation

Measurements from the preceding local checkpoint: Chromium, eight 30-second stereo tracks; test machine and load do not represent the production server. Original reports remain in the project validation directory.

Test	Observed result	Local source
C++ playback	Start 1.239 s; fader response 0.538 s; 0 underruns	r210-continuous-pcm-browser.json
C++ jitter / UI stalls	Network delays up to 180 ms, five 450 ms UI stalls: 0 underruns	r210-continuous-pcm-stress.json
Current engine	Start 2.198 s; cached restart 0.265 s	r210-continuity-current-engine.json



Processing and audio output locations.

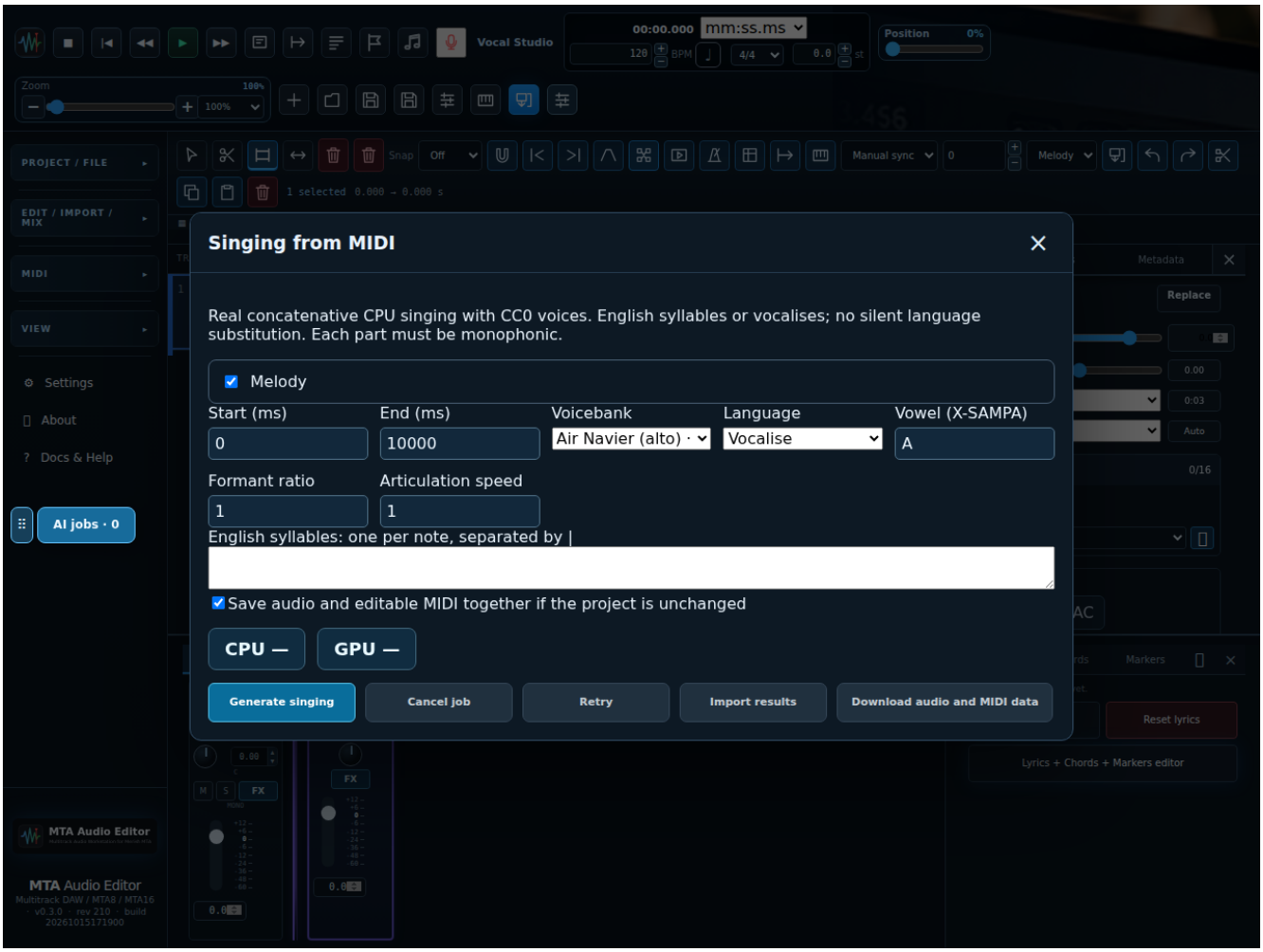
Certification still depends on real-platform checks: audio interface, MIDI driver, third-party plugins, GPU and proxy. Accepted consumption is accounted; source and validity remain in the audit; zero consumption does not prove no GPU use.

GM instruments, singing and licences r210

From AI Studio, open Execution and Studio models to explicitly install the MIT FluidR3 GM bank and CC0 OddVoices voices, with status and progress. OddVoices is CPU concatenative singing, not a neural DiffSinger backend. Singers can render monophonic MIDI parts as vocalises or English syllables, saving audio and MIDI together. Italian/French and voice conversion are not declared available.

Accurate Audio-to-MIDI uses YourMT3 and Basic Pitch, retains original hypotheses and permits importing editable MIDI alternatives. It assigns no false note probability and does not guarantee accuracy for ten arbitrary instruments. Maximum accuracy remains disabled.

Native apps and local processing are free. Any server charges reimburse CPU/GPU/disk infrastructure costs without profit; model licences still apply. Policies, disabled components and manual activation conditions are documented in docs/MUSIC_ASSETS_AND_LICENSES_R210_EN.md in the package. The new verify_music_hardware.py script prepares ASIO/ROCm testing; inventory is not certification.



Singing from MIDI: actual application interface; language follows user preferences.

MIDI projects, musical time and interface zoom

Multitrack DAW + MIDI allows audio and MIDI tracks; **MIDI** allows only MIDI. DAW, MTA8 and MTA16 remain audio-only. Use **Setup** or **Project info → Convert project type**: incompatible tracks are preserved inactive and restored by a compatible conversion. This does not transcribe audio or render instruments.

The first metronome click, including pre-roll, defines the musical project origin. **Return to start** seeks there. The transport shows **mm:ss.ms** or **Bar:Beats:Division:Ticks**, with four divisions per beat and three-digit ticks. Auto uses musical time when BPM is calculated or entered; the initial120 value alone is a placeholder. Fixed, zonal and adaptive tempo share the MIDI grid. MIDI export defaults to the audible project tempo.

In the MIDI editor, **Quantize selected notes** uses the chosen grid, strength and swing and supports undo. Source notes shared by multiple clips use their first occurrence as the timeline reference.

Desktop apps have a top-bar **Interface zoom** control for the whole UI, from50% to200%, separate from timeline zoom. Shortcuts: Ctrl/Cmd +, − and0 to reset. On iPhone/iPad use Options → Interface zoom; on Android use ⋮ → Interface zoom.

Qixuan IT/FR retains contextual French phonemization, including liaison. The bank projects onto existing EN/JA/ZH phones: accent, vowels and pronunciation require listening and review. Installation and tests do not replace the bank and vocoder’s individual terms.

36. Track matching and waveform editing

Select an audio track and choose **Allinea BPM e tonalità** (match BPM and key) in its context menu. A selected timeline range limits the operation; otherwise it applies to the entire track. Select another audio track and run analysis: the dialog proposes source BPM, project zone BPM in source time (before global speed), audible BPM and transposition relative to the reference. Values, including pitch in semitones, can be edited before applying.

Estimates are advisory: listen and correct BPM when necessary. Initial BPM and key estimates use at most 20 seconds of the selection; if a value cannot be estimated, enter it manually. Synchronization tracks source beats across the range and the reference track's beat phase, adapting them to the project BPM map, including zones and adaptive tempo. The initial source BPM can be edited to help choose the correct beat subdivision; destination BPM comes from the project grid. If the reference conflicts with the project map or there are not enough reliable rhythmic attacks, the operation stops: choose a more rhythmic interval or synchronize the reference first. If the reference ends before the result, the remainder follows the project grid. Always listen: syncopation, weak attacks and unrhythmic sections may cause incorrect estimates or artifacts at transitions. Stretching shifts later material on the same track. Committing creates a new source and consolidates the track segments into one clip; original media is retained.

After clicking a track header, use Ctrl+A or Cmd+A to select all tracks, or choose **Select all tracks** from its context menu. After clicking the timeline, the same shortcut selects only segments. Inside text fields it selects text.

For MTA export, **Normalize before export** measures the peak of the summed slots and applies the same gain to each before MP3 encoding. Actual peaks also depend on encoding and playback hardware. Lower the target if an individual slot would clip; active master inserts must be bypassed because they cannot be reproduced proportionally in separate slots.

In **Waveform / sample editor**, choose **Amplifica / attenua (dB)** (amplify / attenuate), set a value from -60 to +24 dB and apply it to the selection. Positive gain may clip. Apply in-place updates the track and waveform; editor Undo and Redo use the project history during the session and preserve the required audio sources.

In **Lyrics + Chords + Markers**, **Delay lyrics (ms)** cumulatively offsets all lyrics; a negative value advances them. Words and syllables follow each line and manual anchors stay attached; chord and marker times are not moved. Changes that would put events outside the project are rejected and can be undone before saving. PDF chords and wrapped lyrics stay together when the block fits on one page; longer text continues on the next page.

37. Timeline segments: selection, split and Join

After clicking the timeline, **Ctrl/Cmd+A** selects all segments independently of sidebar track selection. **Delete/Backspace** removes selected segments. With the **Select** tool, arrows move selected segments; Shift uses a free 10 ms nudge instead of the normal 100 ms or grid step. With no selected segments, arrows move the transport; **Shift+arrows** moves exactly 100 ms. Text fields retain their normal editing shortcuts.

With the **Split** tool, select segments, including all segments, and press **S** or choose **Split selected segments at transport** in the context menu. Splits use the exact transport position, source offsets and track delay. Sidebar tracks do not also need to be selected. Without a segment selection, S operates on selected tracks.

Join, on the horizontal toolbar and timeline context menu, requires exactly two adjacent audio segments on the same track. Gaps and overlaps are not automatically repaired. Consecutive source ranges need only a metadata edit; otherwise MTA prepares a new float WAV source while preserving timing, other clips and original sources. Gain, mixer, pitch and tempo are not applied twice. Undo/Redo restores the project and required sources and refreshes playback audio.

Context menus create empty audio or MIDI tracks according to project type and insert existing project-library audio at the transport. Copy/paste and dragging transfer segments between tracks. Imported lyric phrases of 40 characters or more are split near the middle, preferring punctuation and conjunctions. This is a heuristic, estimating times when synchronized words are unavailable. Editors flag long lines and provide a split command.

38. Shared projects and dashboard

The owner opens **Project sharing**, adds users by username or email and revokes access from the same list. Shared projects are labelled **Shared** in the project picker and can be open simultaneously for participating users.

The separate **Shared dashboard** Sidebar entry appears only while the project is shared. Its window can be dragged, minimized and closed; closing it does not end sharing. A colored badge counts unread messages. Opening the dashboard marks available messages as read; a minimized window resumes displaying updates when restored.

Messages show a bold sender name and a user-specific color. Send text, images and audio, or choose **Record voice** when the browser grants microphone access. Recording requires a secure browser context and a supported format. Email notifications include the project and sender names and exclude the author; they require configured SMTP and active recipients with confirmed email addresses.

Text remains until sharing ends. Image/audio attachments normally expire after **24 hours**, configurable by the administrator; expired files are removed by the server. **Leave sharing** removes your access while leaving your messages for others; the owner can invite you again. Ending sharing also removes the Sidebar entry.

A live server connection delivers shared updates. A session without local edits loads the new version; unsaved local edits instead show a conflict and can be downloaded before reloading. Owner saves take priority for conflicting fields; independent participant edits are retained when reconcilable. A connection loss does not mean a save succeeded.

39. AI benchmark and validation

The administrator opens **AI benchmark and validation** to prepare a corpus, select candidates, start or reopen a campaign, and inspect metrics, outputs and reports. Blind comparison uses Candidate A/B labels, looping excerpts, waveform and volume controls, and a choice between original audio and level-matched listening. Original metric files remain unchanged.

Rating forms save six scores from 1 to 5, comments and a final assessment; for contamination, 5 means negligible interference. Ratings bind to the exact files, checkpoint, pipeline and options. Installation, completed inference, automatic validation and human approval remain separate states.

Current state: Awaiting human validation. The initial corpus generates 12 synthetic examples and 6 negative/difficult cases. The 6 required realistic recordings are not yet available. The registry contains 11 candidates; only Mega 53 has an executable adapter, subject to verified installation and authorization. This delivery does not establish real neural inference or approved professional quality. SAM Audio is no longer available. Initial listening will be organized in groups of 6-10 cases after real model execution.